```
1   PROGRAM PLC_PRG
2   VAR
3       Analog_Input_Raw  :  INT  :=  16384 ;      (* Raw value from flowmeter,
    e.g. %IW64 *)
4       Setpoint_MW200    :  INT ;      (* VFD_Set point *)
5       HMI_Display_Percent  :  REAL ;
6       temp_Real         :  REAL ;
7       scaled_Value      :  REAL ;
8   END_VAR
9
```

```
1   (* STEP 1 & 2: Normalize and Scale *)
2   (* Converting INT to REAL for accurate division *)
3   temp_Real  :=  INT_TO_REAL ( Analog_Input_Raw ) ;
4
5   (* Scaling from 0-27648 to 0-16384 *)
6   scaled_Value  :=  ( temp_Real  /  27648.0 )  *  16384.0 ;
7   HMI_Display_Percent  :=  ( temp_Real  /  27648.0 )  *  100.0 ;
8   (* Convert REAL back to INT *)
9   Setpoint_MW200  :=  REAL_TO_INT ( scaled_Value ) ;
10
11  (* Ensures the final value stays strictly between 0 and 16384 *)
12  Setpoint_MW200  :=  LIMIT ( 0 ,  Setpoint_MW200 ,  16384 ) ;
13
```